Solvency Field Theory V13.14

FORMATION

How Light Enters a Persistent Particle Branch

Electron-positron pair formation as first-turn entry into the maintained recurrent state

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Status
GOOD-ENOUGH V1 WORKING FORMATION MODEL. The maintained electron is already frozen at V13.12. V13.14 proposes the simplest formation path consistent with the earned formation and maintenance results. It is not a zero-free-parameter or first-principles pair-production theorem.

Abstract

SFT V13.12 established a parameterized, reaction-completed, recurrent electron state that remains stable under its maintained support/geometry dynamics. V13.14 asks the complementary question: how can two open photon-like carriers enter that persistent branch? The V1 answer is intentionally minimal. In QFT/SM the process is called two-photon electron-positron pair production; in native SFT language the carrier stock does not convert into a different substance. The same locally null carrier stock is reorganized into two recurrent, charge-conjugate structures.

The proposed formation sequence has two entrance conditions. First, the two-null invariant energy must make the electron/positron branch kinematically accessible. Second, the joint event must generate a real first inward turn together with the nascent support/reaction state. The formation program has independently demonstrated a connected two-photon response, relational steering data, mirror inward first turns on two outgoing carrier sectors, a concurrent scalar support mode, and a rotating trace-free reaction saddle. That saddle is the same geometric operator class used by the recurrent optical stability theorem underlying V13.12.

At reduced mean-route scope, the first-loop obligation can be represented with one explicit formation-to-recurrence normalization: the measured formation first-turn readout maps into the independently earned exact-tangent-source circular turning band while centered-offset, co-propagating, and large-offset controls remain suppressed. This is therefore classified PASS WITH FREE PARAMETER, not derived. Once exact same-support recurrence occurs, the already-existing reaction becomes cycle-owned without a physical handoff, and V13.12 supplies the maintained regime. Above threshold, surplus invariant energy changes daughter relative motion rather than continuously changing the intrinsic electron branch.

The remaining V1 opens are explicit: the absolute support normalization Gamma_star, the first-turn-to-sustained-loop normalization, dynamic birth of the charge/framing label, and a complete time-dependent two-photon-to-two-recurrent-particle execution. V13.14 therefore presents a developable formation mechanism rather than a completed microscopic formation theorem.

The picture in plain language

Light does not turn into matter. In this V1 picture, it gets caught. The same locally null carrier stock that was propagating openly is reorganized into two closed, self-maintaining structures. Light going somewhere becomes carrier stock going around. Formation is not a change of substance; it is a change of branch.

At the entrance level resolved by V13.14, the story reduces to two gates. The first is a price of admission: the electron-positron branch is kinematically accessible only when the two-photon invariant energy satisfies the pair threshold, corresponding to 1.022 MeV of center-of-momentum energy for the electron branch. Below that threshold this branch cannot form. Above it, the intrinsic electron/positron branch does not continuously grow heavier; in the ideal two-body channel the surplus appears as opposite daughter motion, while radiation or other open channels may share the surplus when allowed.

The second gate is a turn. Two beams passing each other do not become loops under source-free Maxwell evolution. The declared common SFT response must make a real inward turn. The formation engine does: in the offset-steered diagnostic channel, two separately logged outgoing carrier sectors acquire mirror inward tangent shifts without inserting an electron route or V13.12 target. How much turn is available is controlled, at leading diagnostic level, by how strongly the packets overlap and by whether the encounter carries a datum that can orient the response. Exact centering has maximum scalar overlap but no offset-handed orbital direction; a large miss has a lever arm but almost no overlap. Other centered steering or framing channels, including polarization/helicity structure, remain possible and are not replaced by the offset channel.

The part that makes the picture cohere is that the local machinery that keeps a mature electron stable is already present during formation. The event produces a nascent support/reaction state together with a rotating trace-free saddle of the same geometric class used by the recurrent maintenance theorem. The collision therefore does not make a raw loop and hand it to a second machine. There is only one evolving carrier/support/geometry system.

The remaining seam is the first complete same-support return of the full state - carrier, support/reaction, phase, and framing - not merely a spatial revisit. V1 does not yet compute that return from scratch in one uninterrupted time-dependent solve. Instead it exposes one explicit first-turn-to-loop normalization and gives the tested range required to connect the physical first-turn readout to the independently earned exact-tangent-source loop class. Exact full-state return changes cycle ownership; it does not transfer the carrier into a new mechanism.

Two other debts remain visible. V13.12 can maintain a supplied absolute support normalization but does not select its formation value, and V13.14 does not yet dynamically derive the signed charge/framing labels of the conjugate daughters. The paper also does not compute a pair-production cross section or rate, and it does not derive the electron mass or charge magnitude from the formation calculation.

The model is deliberately falsifiable. A one-photon state, co-propagating photons, a sufficiently wide miss, or insufficient invariant energy must not acquire false recurrent capture. And when the first loop is eventually evolved without a circular scaffold, it must close as two local full-state returns rather than one crossed combined object.

So the picture in one breath is: one light carrier is turned by another strongly enough to enter a recurrent branch that is already energetically available, using the same support and rotating geometry that later maintains the particle; once full recurrence exists, the state keeps itself, while surplus energy goes into motion or other allowed channels rather than into a different electron.

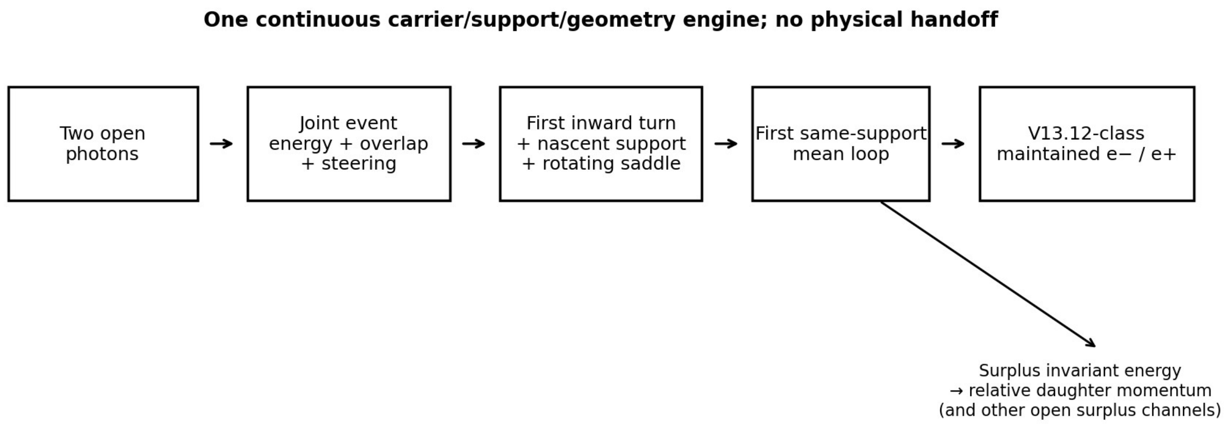

Figure 1. V13.14 formation architecture. The same carrier/support/geometry engine spans formation and maintenance.

1. Scope and claim boundary

This paper is a V1 formation model for the electron-positron branch. Its purpose is to state the smallest mechanism supported by the current SFT formation program and the frozen maintained-electron result. It does not reproduce the full multi-day derivation history, failed engines, or every diagnostic gate. Those remain in the internal provenance stack.

The central claim is narrower: the physics needed locally after the first turn is not a new mechanism. Formation already produces the same kind of rotating trace-free support geometry used by the maintained recurrent electron. The remaining seam is mean-route recurrence and its absolute normalization.

Public claim boundary
V13.14 does not claim a zero-parameter electron mass derivation, a derived pair-production cross section, a derived absolute support radius, a derived value of alpha or $|q|$, or a complete time-dependent two-photon-to-two-particle simulation. Where the V1 mechanism uses an interface parameter, it is named explicitly.

2. The physical object: open and recurrent organization of one carrier stock

SFT uses one carrier ontology across the process. Locally, the carrier remains photon-like and null. The difference between light and matter is organizational: an open carrier continues outward, while a massive particle is a persistent recurrent structure made from the same carrier stock. Pair formation is therefore not modeled as photon substance converting into a new material substance.

The final state must contain two independently recurrent structures. The incoming photons are not preassigned one-to-one to the electron and positron. During the encounter they contribute to one common physical configuration; after separation, the two emergent carrier sectors must each acquire their own same-support recurrence, reaction/support state, local clock, framing label, and momentum.

3. Starting point: what V13.12 already owns

V13.12 is a parameterized stable maintained electron, not a formation theory. Its frozen representative uses support normalization Gamma_star = 0.18562349764677286 and support ratio s = 0.24965573857744025. Those are supplied maintenance inputs; Gamma_star is not promoted as a first-principles prediction of the absolute electron radius.

The maintained background is one finite tangent-null carrier source with an exact transverse G12U profile f_perp and a distinct, more centrally concentrated quadratic settlement margin q_perp. The complete state also includes lawful support reaction, electromagnetic dressing, self-consistent geometry, and the physical transported frame. The occupied carrier follows the geometry generated by that complete state.

Its transverse stability is elliptic rather than dissipative. The frozen state has a/nu^2 = 0.0864525425 inside the strict interval 0 < a/nu^2 < 1. Fresh Floquet multipliers lie on the unit circle to numerical precision, a strictly positive invariant M^T G M = G exists, and direct powers remain bounded through 100,000 cycles. A free-geodesic negative control misses the nominal route badly, showing that the particle is the complete recurrent support/metric/current structure rather than a passive photon on a static racetrack.

$$0 < a / nu^2 < 1$$

V13.12 also establishes a broad maintenance family. The support audit remains elliptic over at least s = 0.05 through s = 0.60, while the optical stability ratio changes only weakly. The mature maintenance action can hold a supplied support width and ellipticity, but by construction it does not select why nature chose that absolute width in formation.

4. Formation reduced to two entrance gates

4.1 Invariant-energy accessibility

For two incoming real photons with energies E1 and E2 and relative angle theta, the invariant energy is

$$s = 2\ E1\ E2\ (1 - \cos(\theta)).$$

The electron-positron branch is kinematically accessible when

$$s >= (2\ m_e\ c^2)^2 = 4\ m_e^2\ c^4.$$

In the center-of-momentum frame, once that branch is selected, each daughter has energy sqrt(s)/2 and momentum magnitude

$$p_*\ c = \mathrm{sqrt}(\ s/4 - m_e^2\ c^4\ ).$$

This gives the first major simplification: collision energy does not continuously set a different intrinsic electron mass. At threshold the relative daughter momentum vanishes in the ideal two-body channel. Above threshold, the same intrinsic branch remains available while the surplus appears as relative daughter motion. If additional radiative or particle channels are open, they may share the surplus.

4.2 Geometric entry: compounding plus a turning datum

Energy accessibility alone does not decide whether the encounter captures into the recurrent branch. The formation event also needs enough common response and a physical turning orientation. The current SFT formation program separates those jobs cleanly: overlap and collision angle control the common compounding strength, while impact parameter can provide an orbital lever arm. Polarization/helicity can supply an independent framing or handed seed, so a nonzero impact parameter is not claimed to be the universal pair-production channel.

Head-on scalar-compounding theorem. For two exactly counterpropagating null carriers, with k_hat_2 = -k_hat_1, transverse electric fields E_1 and E_2, B_1 = k_hat_1 x E_1, and B_2 = -k_hat_1 x E_2, the two cross Lorentz invariants reduce, up to fixed sign/factor conventions, to

F_1 . F_2 proportional to -2 (E_1 . E_2), F_1 . *F_2 proportional to 2 k_hat_1 . (E_1 x E_2).

Because each isolated carrier is null, the R3A7 root-free null-departure density of the joint field is

E_inv = (F_1 . F_2)^2 + (F_1 . *F_2)^2 proportional to 4 |E_1|^2 |E_2|^2.

Using (a . b)^2 + |a x b|^2 = |a|^2 |b|^2, the exact head-on null limit shows that, at fixed local field magnitudes, this scalar compounding channel is independent of the relative transverse polarization angle. This is a narrow separation theorem: polarization cannot modify this even scalar factor, although polarization can still enter formation through other signed spin-two, framing/handedness, or finite-packet/non-head-on response channels.

For identical translated packets, integrating this scalar makes the idealized overlap factor a normalized cross-correlation of the local intensity profile I = |E|^2. For a pure Gaussian amplitude envelope A(r) proportional to exp[-r^2/(2 w^2)], the corresponding ideal overlap is

O(b) = exp[-b^2/(2 w^2)].

The Gaussian reference coefficient is therefore 1/2 when the dimensionless offset is x = b/w. The actual ENG1C/R3 production packet is not a literal real-space Gaussian: its analytic spectrum uses a smooth compact bump multiplied by a Gaussian spectral envelope. The fitted coefficient 0.452792 below is therefore retained as a packet-specific diagnostic compression, not promoted to a new constant or converted into a physical width without a direct autocorrelation audit using the frozen packet definition.

For the offset-steered diagnostic channel, the independently measured finite-packet compounding fraction is very nearly a one-parameter Gaussian in x = b/sigma:

O(x) ~= exp(-0.452792 x^2).

Combining that even overlap strength with the odd steering lever arm gives the simplest first-turn capacity

T(x) proportional to x exp(-0.452792 x^2).

This compression is diagnostic, not a fundamental new law. It reproduces the previously earned nonlinear impact-parameter turn morphology at R^2 ~= 0.989. It also explains, without adding a mechanism, why exact centering has maximum overlap but no offset-handed steering, large miss distance has a lever arm but negligible overlap, and an intermediate offset is best for this particular channel.

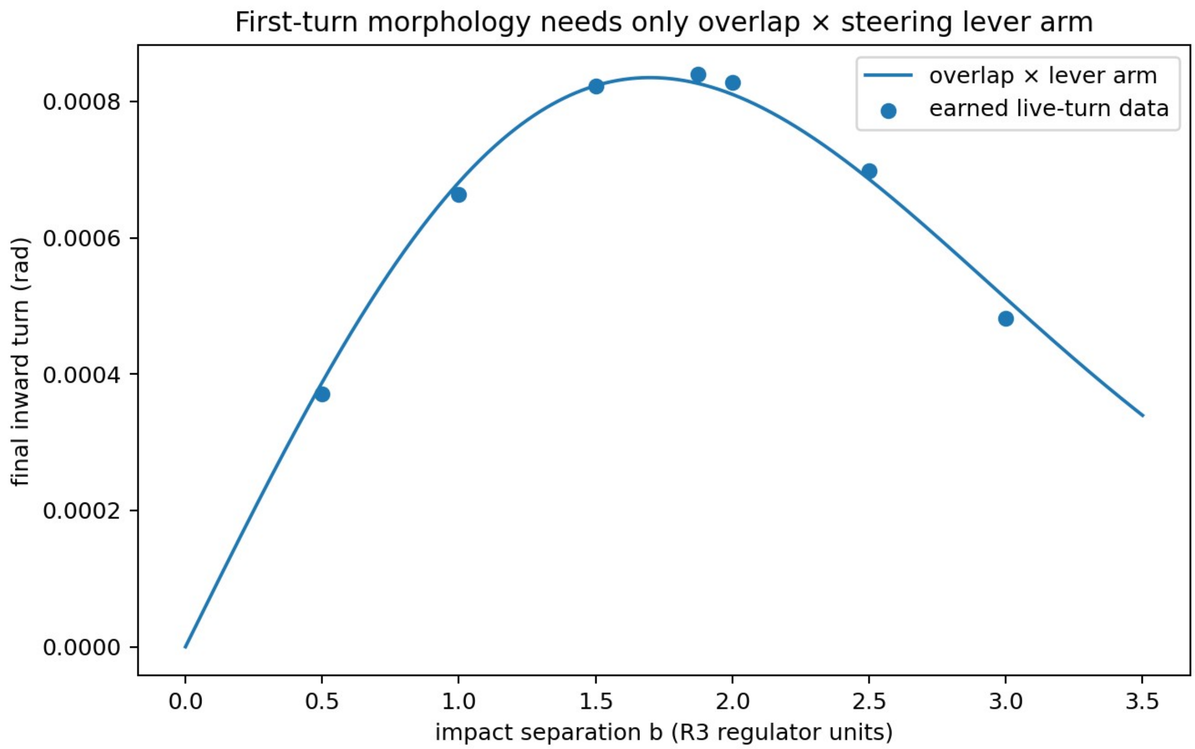

Figure 2. Diagnostic compression of the offset-steered first-turn morphology: overlap x lever arm. The regulator scale is not a physical prediction.

5. The event produces a real first turn

The load-bearing formation result is not merely a kinematic angular-momentum seed. The constrained common response has been lifted into the physical carrier variables. In the baseline offset event, the two outgoing carrier sectors are separately loggable and receive mirror inward tangent shifts of approximately

$$\text{theta\_R} = 0.01643321 \text{ rad}, \quad \text{theta\_L} = 0.01633254 \text{ rad}.$$

These angles belong to the regulator realization used by the formation engine and are not proposed as universal pair-production observables. Their role is mechanistic: the common two-photon response can produce a real physical first inward turn while preserving the isolated-photon, co-propagating, separated, large-offset, phase, and refinement controls.

At the same time, the two outgoing sectors expose nonzero nascent support/reaction variables. Importantly, daughter-local retained settlement/self-reaction is allowed to begin while the joint formation source is still nonzero. The reaction state does not wait for an artificial switch into a second 'maintenance engine.'

6. Formation already contains the maintenance geometry class

The strongest structural result in the current program is that the formation reaction is already a rotating trace-free saddle. In the physical screen, the constrained reaction tensor can be written

$$S\_TF(Y2) = [[Re(Y2), Im(Y2)], [Im(Y2), -Re(Y2)]].$$

For every nonzero complex Y2 this is identically

$$S\_TF = |Y2|\, R(psi)\, diag(1,-1)\, R(psi)^T, \quad psi = arg(Y2)/2.$$

Its principal values are +|Y2| and -|Y2|. The headless principal-axis direction is psi = arg(Y2)/2. The formation source phase winds through more than nine complex turns during the active event window, so the event does not deliver one static inward shove; it supplies a rapidly rotating signed shear instruction together with a phase-neutral scalar G12U housing response.

C17A's recurrent optical theorem uses the same geometric object class:

$$K(theta) = a\, R(nu\ theta)\, diag(1,-1)\, R(nu\ theta)^T.$$

In the maintained recurrent state, the physical support axes co-rotate and the saddle-plus-half-twist system is elliptic when 0 < a < nu^2. No new closed-only position-space kinetic operator is required by that theorem. The local branch machinery present during formation is therefore of the same type as the machinery that maintains the electron.

Interpretation
The V1 formation problem is no longer “find a second force strong enough to bend a photon through 360 degrees.” After the first turn, formation already contains the local support, scalar housing response, trace-free saddle, and rotating headless frame needed by the recurrent maintenance class. The remaining seam is the first completed same-support mean route.

7. The first same-support loop

7.1 Independent exact-tangent-source turning control

C17D independently asks whether the exact finite G12U tangent-null electron source can sustain a circular mean route in a reduced finite-torus tensor/binding control. Across the tested support family, the required reduced mean-route turning coefficient lies in

$$0.1057234868 <= C\_required <= 0.3480058363,$$

while the exterior compactness control remains below one half:

$$0.1009796275 <= C\_A <= 0.2960307130 < 1/2.$$

This is a positive mean-route turning and compactness result for the exact distributed source. It is not a formation derivation because C17D begins from an already circular torus, and it is not the full nonaxisymmetric Einstein-worldtube root.

7.2 One explicit V1 interface

The physical first-turn angle produced by the formation engine and the sustained turning coefficient used by C17D are not canonically normalized to each other. V1 therefore exposes, rather than hides, one cross-normalization:

C_loop = eta_(turn->loop) * theta_first.

Using the mean physical first-turn readout theta_first = 0.0163828762 rad, the entire tested C17D legal-loop band is reached for

6.4533 <= eta_(turn->loop) <= 21.2420.

This interval is not a prediction of a new constant. It is the explicit price of connecting two independently normalized reduced engines. The result is nevertheless useful because the same minimum normalization that places the main offset event at the easiest legal loop edge leaves the centered offset-steered channel at zero, the large-offset control at only about 0.7% of that edge, and the co-propagating control at about 1.2 x 10^-7 of it.

V1 ruling
FIRST SAME-SUPPORT MEAN LOOP: PASS WITH FREE PARAMETER at reduced exact-tangent-source scope. The free quantity is eta_(turn->loop). A full forward time-dependent two-photon-to-worldtube loop execution remains open.

8. No physical handoff at recurrence

The first exact same-support return changes a topological classification, not the substance or underlying engine. Before exact return, a daughter-local nascent reaction may already be nonzero. At exact full-state boundary cancellation, that already-existing reaction becomes cycle-owned. B1C2 found zero reaction-state discontinuity at this ownership change in its structural test.

The intended chronology is therefore continuous:

joint event -> daughter-local nascent reaction -> exact local return -> cycle ownership -> maintained recurrence.

No reset, projection, carrier replacement, or post-formation switch is introduced. As the daughters separate, the shared two-photon compounding naturally fades; if the local recurrent support/geometry state has formed, the state simply remains in the V13.12-class maintenance regime.

9. Surplus energy and daughter motion

The branch-surplus separation is part of the simplicity of the model. Once a stable electron/positron branch is selected, additional invariant energy does not continuously 'fatten' that root. In the isolated two-body center-of-momentum channel it appears as opposite daughter kinetic momentum. More generally, radiation or additional particle channels may absorb surplus if they are open.

This provides a direct physical reading of above-threshold pair production: the internal recurrent branch stays the electron/positron branch while extra collision energy primarily changes how the two finished structures move relative to one another.

10. V1 status ledger

Object	V1 status	Meaning
Two-null branch accessibility	**OWNED**	s >= 4 m_e^2 c^4; above threshold surplus becomes relative motion.
Connected two-photon compounding	**OWNED**	Common-null and separated controls suppress the response.
Turning orientation data	**OWNED**	Impact parameter is one odd steering channel; helicity/polarization may supply others.
Physical first inward turn	**OWNED**	Mirror inward tangent response on two outgoing carrier sectors.
Nascent daughter support/reaction	**OWNED STRUCTURALLY**	May begin before exact recurrence; no handoff is required.
Rotating trace-free formation saddle	**OWNED**	Same 2D saddle geometry class used by recurrent optical maintenance.
Maintained electron structure and stability	**OWNED**	Frozen V13.12 maintained Electron V1.
Absolute support normalization Gamma_star	**PASS WITH FREE PARAMETER**	Maintenance holds a supplied width but does not select it.
First turn -> sustained first mean loop	**PASS WITH FREE PARAMETER**	Reduced exact-tangent-source scope via eta_(turn->loop).
Charge/framing birth	**PLAUSIBLE / OPEN**	Opposite conjugate framing is not yet dynamically derived by the complete formation solve.
Full forward pair formation	**OPEN**	No complete time-dependent two-photon -> two reaction-completed recurrent-particle execution yet.
Pair-production rate / cross section	**NOT CLAIMED**	No scattering-rate prediction is made in V1.

11. What V13.14 does not yet derive

- The absolute formation/support normalization Gamma_star. V13.12 can maintain a supplied value but does not select it from mature maintenance dynamics.
- The cross-normalization eta_(turn->loop) that converts the transient first-turn readout into the reduced sustained-loop turning normalization.
- The complete dynamic birth of the signed charge/framing label and its exact relation to the two conjugate daughters.
- A single full time-dependent calculation that starts from two photons and ends with two independently recurrent V13.12-class objects without any reduced circular scaffold.
- A pair-production probability, differential cross section, polarization-resolved rate, or absolute event rate.
- A first-principles derivation of alpha_EM, the absolute electric charge magnitude, or the electron mass from this formation model.

These are not hidden failures. They define the V1 research boundary. In particular, a parameterized first-loop interface is acceptable for the working model but cannot be promoted into a universal constant or prediction until its normalization is derived from the common action.

12. Development and falsification targets

The V1 model is useful because its remaining obligations are narrow enough to test directly. The next work should not add new formation fields or forces unless the existing common-action structure fails one of these tests.

1. Derive or eliminate eta_(turn->loop) by placing the formation first-turn and recurrent turning control in one canonical normalization.
2. Execute the first same-support mean loop dynamically from the formation state without inserting a circular route.
3. Demonstrate two local returns rather than a crossed single combined cycle.
4. Derive the conjugate framing/charge labels from the same event while preserving equal mass-like readouts.
5. Verify that one-photon, co-propagating, large-offset, and insufficient-invariant-energy controls never acquire false recurrent capture.
6. Only after these pass, attempt a cross section or rate calculation.

Stop rule
If a complete common-action evolution cannot connect the earned first-turn/support state to a lawful recurrent mean route without adding a new ad hoc force, the V1 formation mechanism must remain parameterized or be revised. Do not rescue it by hiding a fitted route or a new post-formation machine.

13. Conclusion

SFT V13.14 proposes that electron-positron formation is branch entry, not substance conversion. The two-photon event first has to make the electron/positron branch kinematically accessible. Its common overlap and relational steering then generate the first physical inward turn. The same event already produces a nascent support/reaction state, a concurrent scalar housing response, and a rotating trace-free saddle of the same geometric class that stabilizes the maintained electron.

At reduced V1 scope, one explicit free normalization is sufficient to connect the actual formation first-turn readout to the independently earned exact-tangent-source first-loop turning class while preserving the decisive null and separation controls. Exact recurrence then makes the already-existing daughter-local reaction cycle-owned; there is no physical handoff into another machine. V13.12 supplies the subsequent maintained regime.

The result is therefore deliberately graded as a good-enough V1 working formation model: the start, local mechanism, first turn, recurrent maintenance class, and surplus-energy bookkeeping are all independently supported; the remaining free interfaces and open dynamics are named. The model is ready to be developed and attacked as a compact formation program rather than rebuilt as another large mechanism stack.

Appendix A. Core equation spine

Object	Equation
Two-null invariant energy	s = 2 E1 E2 (1 - cos(theta))
Electron/positron accessibility	s >= 4 m_e^2 c^4

Object	Equation
CM daughter momentum	p_* c = sqrt(s/4 - m_e^2 c^4)
ENG1C packet-fit overlap diagnostic	O(x) ~= exp(-0.452792 x^2), x = b/sigma (packet-specific compression)
Offset-channel turn-capacity diagnostic	T(x) proportional to x O(x)
Formation trace-free saddle	S_TF = |Y2| R(psi) diag(1,-1) R(psi)^T
Reaction frame	psi = arg(Y2)/2
Recurrent optical saddle	K(theta) = a R(nu theta) diag(1,-1) R(nu theta)^T
Elliptic maintenance condition	0 < a < nu^2
V1 first-loop interface	C_loop = eta_(turn->loop) theta_first
Head-on scalar compounding	E_inv proportional to |E_1|^2 |E_2|^2; relative polarization-angle blind in the exact head-on null limit

References and internal provenance

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[2] IEMM-C17G16 — Final Stability Audit of the Frozen Electron V1.

[3] IEMM-C17G4 — V1 Retained-Action / Bregman Executor, Gamma Maintenance, and Selector Audit.

[4] IEMM-C17A — Recurrent-Support Optical Invariant and Carrier-Operator Provenance.

[5] IEMM-C17D — Background/Deviation Separation, Exact G12U Tangent Source, and Liouville Nonuniqueness Audit.

[6] UNI-G13L2E — Electron/Positron Pair Formation, Continuous 4pi Closure, Scale Orbit, and Eigenvalue Audit.

[7] F1-ENG1C — Relational Offset and Maxwell Ablation.

[8] F1-UNIFIED1-B1B — Constraint-Complete Slaved Two-Mode Response.

[9] F1-UNIFIED1-B1C — Two-Entity Physical Carrier Predefect Lift.

[10] F1-UNIFIED1-B1C2 — Two-Daughter Cycle Ownership Onset.

[11] F1-UNIFIED1-B1C3A — Formation-Slaved Double-Return Accessibility.

[12] SFT Pair Formation — Good-Enough V1 Integration Audit v0.1.

[13] SFT First-Turn -> Recurrent-Branch Audit v0.1.

[14] SFT First Mean Loop — Physical First-Turn Interface Audit v0.2.